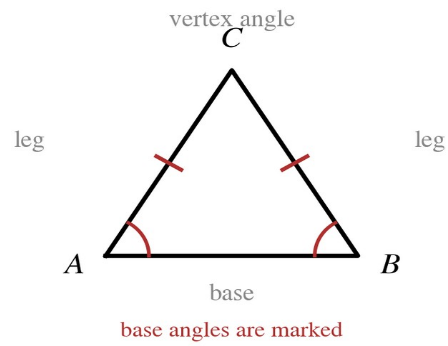
Speak **L**ike **A** **M**athematician:

<u>legs (of an isosceles triangle):</u>

<u>vertex angle:</u>

<u>base:</u>

<u>base angles:</u>



<u>Isosceles and Equilateral Triangle Theorems</u>

Base Angles Theorem: if two ___________ of a triangle are congruent, then the ___________ opposite them are congruent.

Converse of the Base Angles Theorem: if two ___________ of a triangle are congruent, then the ___________ opposite them are congruent.

Corollary: a triangle is equilateral if and only if it is __________________, and each of its angles measures ___________.

Example 1 – Name the Parts

Use the isosceles triangle at the right.

a) Name the legs, the base, and the vertex angle.

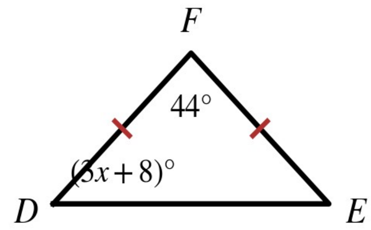
b) Name the base angles. What does the Base Angles Theorem say about them?

Example 2 – Use the Base Angles Theorem

In △DEF, the legs are marked congruent and the vertex angle measures 44°.

a) Find the measure of each base angle.

b) Write and solve an equation for x.



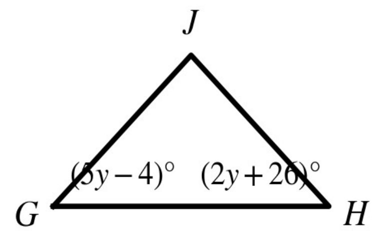
Example 3 – Use the Converse

In △GHJ, the two base angles are marked with algebraic expressions.

a) Write and solve an equation for y.

b) Find the measure of each base angle and of the vertex angle.

c) What does the Converse let you conclude about the sides?

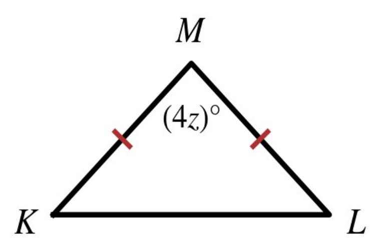


Example 4 – Equilateral and Equiangular

△KLM is equilateral.

a) Find z.

b) Explain why an equilateral triangle must be equiangular.



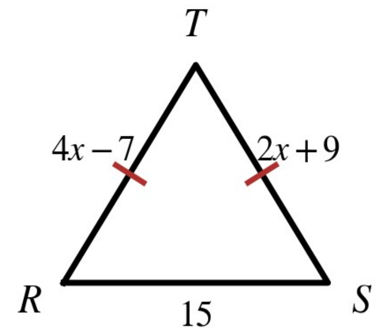
Example 5 – Find x from the Sides

In △RST, the two legs are marked congruent.

a) Write and solve an equation for x.

b) Find the length of each leg. Then find the perimeter.

c) Could the base ever be longer than 50? Explain.



Example 6 – Two Isosceles Triangles

In the diagram, $\overline{AB} \cong \overline{AC}$, $\overline{BD} \cong \overline{BC}$, and m∠A = 38°. Find m∠DBC.

a) Find m∠ABC and m∠ACB.

b) Use △BDC. What are m∠BDC and m∠BCD?

c) Find m∠DBC.